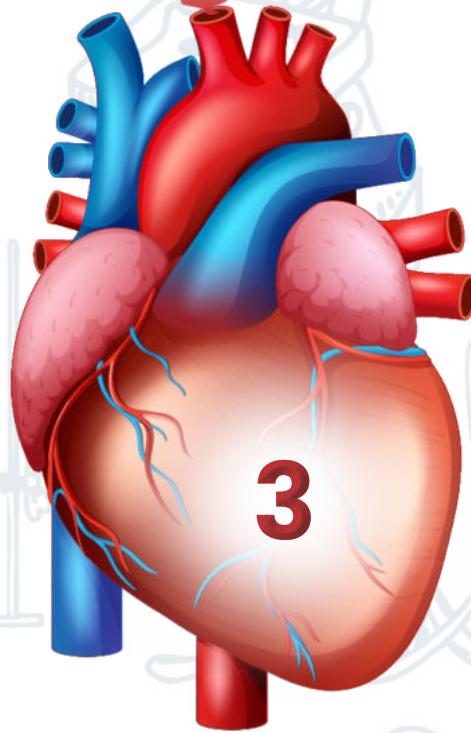
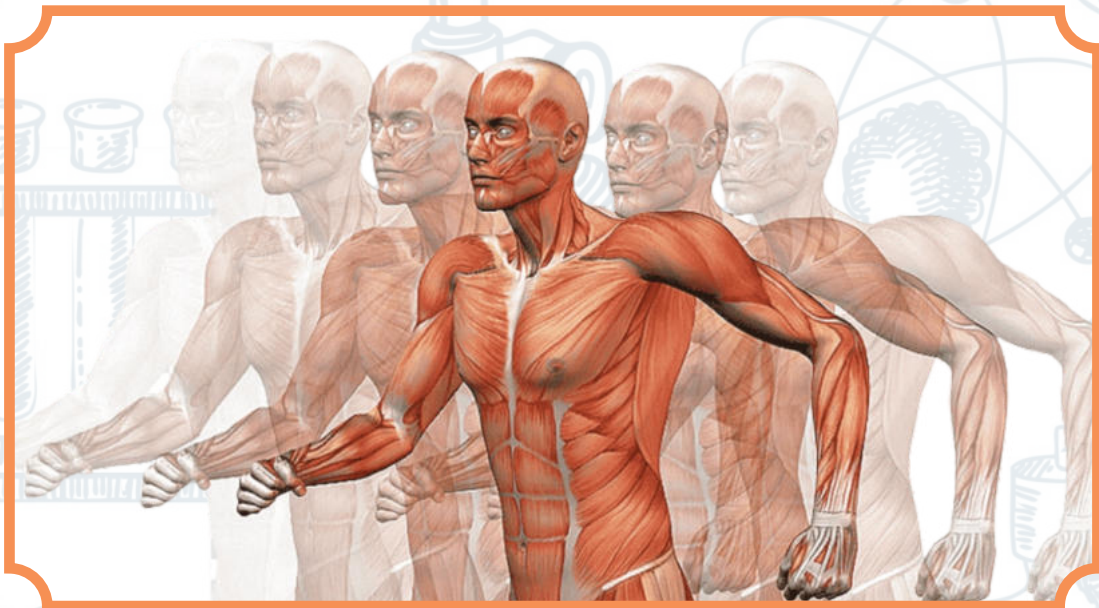


UNIT



CELL STRUCTURE & FUNCTION



CELL FRACTIONATION

Cell fractionation is the combination of various methods used to separate a cell organelle and components based upon size and density. It is very useful for electron microscopy of cell component

Principle

The separation of cell organelles is based on difference in size, density, charge, mass etc.

Method

Cell fractionation consists of two steps i.e. homogenization and centrifugation

1. Homogenization

It is the formation of homogenous mass of cells called cell homogenate or cell suspension.

Process: It involves the grinding of cells in a suitable medium with the presence of certain enzymes that can break the cementing substance of cells. For example, pectinase which digests middle lamella among plant cells.

Procedure

This can be done in a cell homogenizer e.g. food mixer/ blender.

Result

This procedure gives rise to a uniform mixture of cells called cell homogenate.

2. Sedimentation

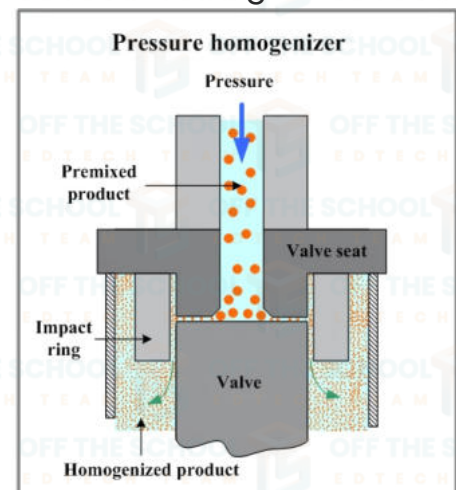
Sedimentation is the process of setting down cell organelles on the basis of density and mass by centrifugation. Centrifugation is the process to separate substances on the basis of their size and densities under the influence of centrifugal force by a machine called centrifuge.

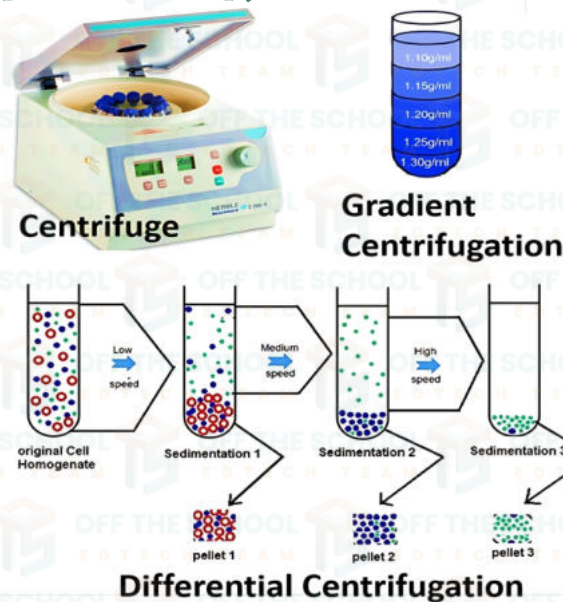
Principle of Centrifuge

- Centrifuge spins the tubes.
- Contents are kept in tubes that are much like the test tubes.
- Spinning the tubes exerts a centrifugal (spinning) force on the contents.
- Contents are separated based on size and density under the influence of centrifugal force.

Procedure

There are two major ways of centrifugations i.e. density gradient centrifugation and differential centrifugation.





Density Gradient Centrifugation

It is based on different densities of cell components

- Cell components of different densities are separated in different layers (sediments) in the tube containing ionic medium according to their size and densities.
- Upper sediments have smaller and less dense components than lower sediments.

Differential Centrifugation

It is based on different sizes and shapes of cell components

- Sedimentation rate for a particle of given size and shape measure how fast the particle settles or sediments.
- The faster the rotation of centrifuge, the smaller the particles will sediment.
- A series of increasing speeds can be used.
- At each step, the content which settles at the bottom of the tube is called pellet.
- At each step, the content which remains suspended above the sediment in the form of liquid is called supernatant.
- After each speed, the supernatant can be drawn off and centrifuge again at higher speed.
- A series of pellets containing cell organelles of smaller and smaller size can be obtained.

original Cell Homogenate

From heaviest to lightest, the order of mass of these organelles is usually;

- Nuclei
- Chloroplasts (if plant tissue)
- Mitochondria
- Lysosomes
- Endoplasmic reticulum
- Ribosomes

DIFFERENTIAL STAINING

Most biological structures are transparent. In order to differentiate between these structures various colour dyes are applied. Such techniques are called staining techniques.

Principle

Different cell components have different chemical composition therefore, they combine with their dyes. Staining of one organelle does not affect other parts.

Stains: Colour dyes used to differentiate or locate cell components are called stains.

Types of Staining: Two main types:

a) Single Staining

• When only one stain is used it is called single staining. • Use of only borax carmine to stain nucleus.

b) Differential Staining / Double Staining

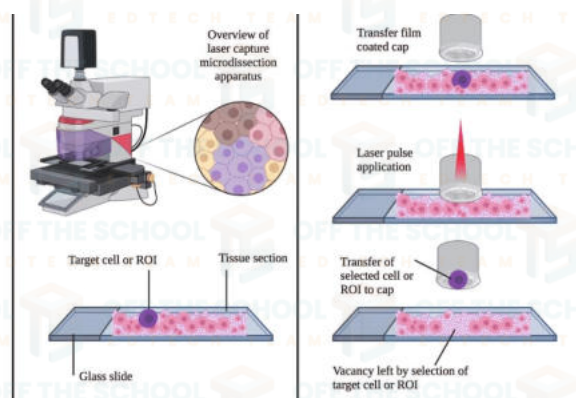
• When two stains are used to stain different parts of cell it is called differential staining

MICRODISSECTIONS

Microdissection refers to the variety of techniques where a microscope is used to assist in dissection.

Principle

Microscope is used to observe very small objects e.g. cells which are not visible to naked eye. Therefore, use of microscope makes the process of dissection for very small objects possible.



Applications

- It is done to remove tumour or granules from delicate tissue or cells like, brain, heart, and nerve cells.
- It is used to obtain pure sample of cells, chromosomes, DNA, proteins etc.
- It is widely used in biological research.

Procedure: In this technique, the image is seen on large TV screen or monitor while dissecting.

TISSUE CULTURE

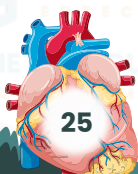
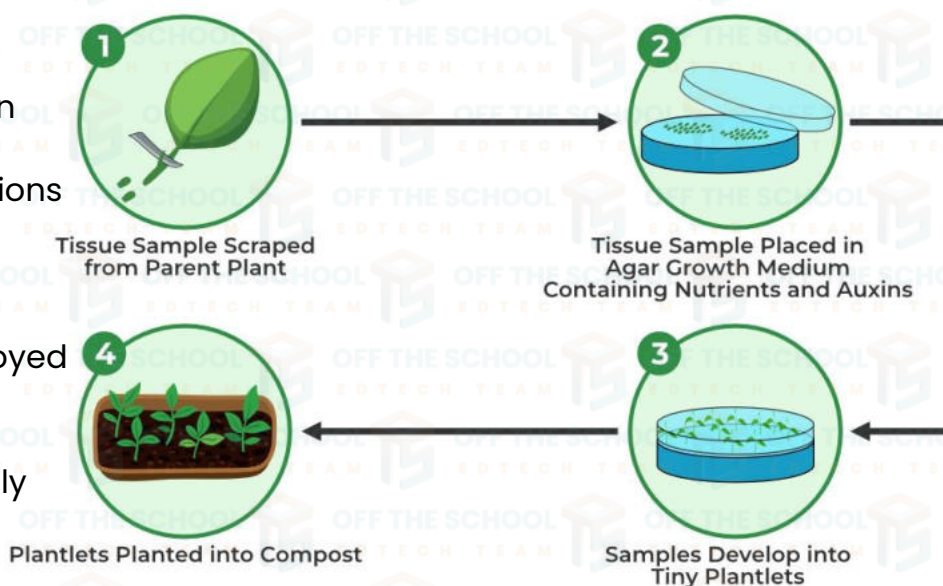
Growth of a cell or a tissue on chemically defined nutrient medium under sterile conditions is called tissue culture.

Application

This technique can be employed for both plants and animals.

a) Plant Tissue Culturing

Plant tissue culturing is mainly used for plant cloning i.e., production of genetically identical plants (clones).



b) Animal Tissue Culturing

Animal tissue culture is usually set up by growing individual cells to form a single layer of cells over the surface of a glass container. Animal tissue cultures are used to see any abnormality in the cell, e.g., cancer, chromosomal disorder etc

CHROMATOGRAPHY

Chromatography is a technique which is used to separate different chemical compounds from a mixture using an apparatus called chromatography chamber.

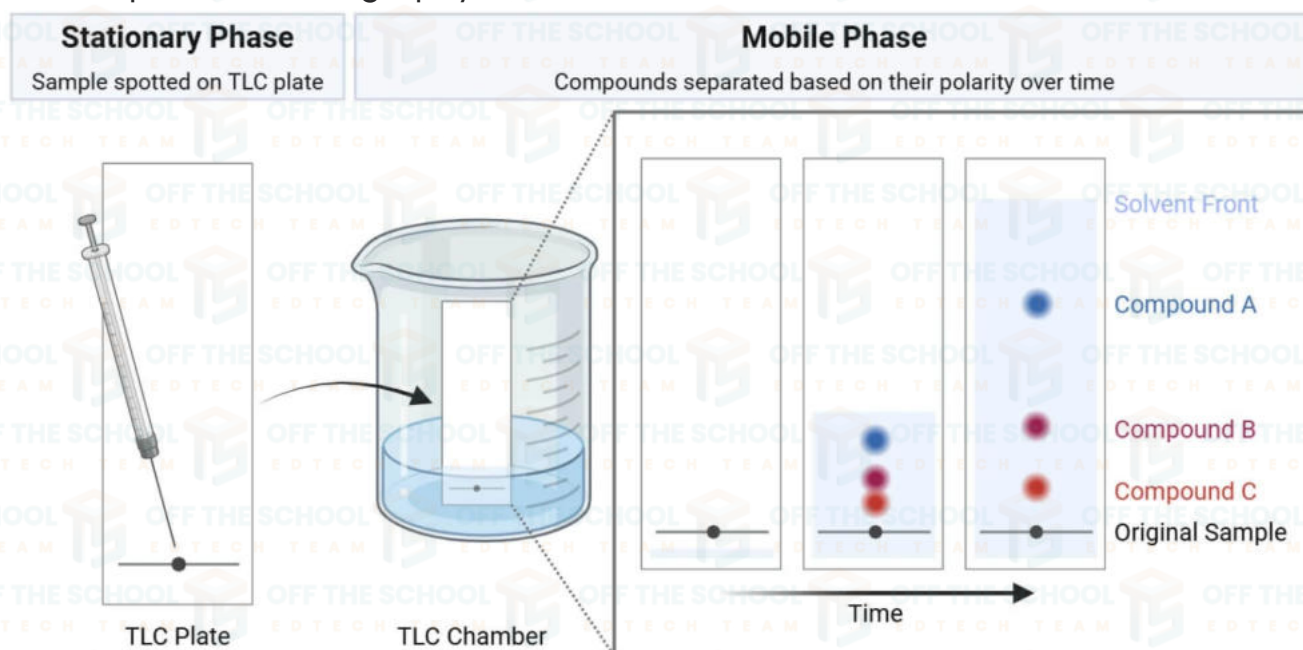
Applications

It is generally used for the separation of mixtures of proteins, amino acids or photosynthetic compounds. [Chrome = color, graphy = to write, originally used for separation of colored compounds]

Types

There are different types e.g.

- Paper Chromatography
- Column Chromatography
- Gas Liquid Chromatography



Paper Chromatography

- It is the most simple
- Most widely used
- It uses a filter paper (that is why it is called paper chromatography)
- It involves two phases i.e. Mobile phase and Stationary phase.

a) Mobile Phase

- It travels over stationary phase (so called mobile phase)
- It consists of mixture dissolved in a solvent e.g. water.

Mobile phase = mixture + solvent

b) Stationary Phase

➤ It is a filter paper. It stays fixed (so called stationary phase) working/ Procedure / Mechanism

➤ When mobile phase travels through stationary phase, components of mixture begin to separate in the form of dots.

➤ Different components move with different speeds so get separated.

It is a slow process.

➤ After due time, paper is let to dry.

➤ Colorless components are made visible by using dyes or stains.

➤ The paper showing results of chromatography is called chromatogram.

ELECTROPHORESIS

• Separation technique.

• Used to separate fragments of charge bearing polymer molecules.

• e.g. separation of DNA, RNA, protein etc.

Basis

It is based on;

Charge i.e. +ve or -ve • Shape • Size • Molecular weight.

Applications

Electrophoresis technique is used for;

• Separation of DNA, RNA, proteins from mixtures.

• Visualizing DNA molecules and determining their approximate sizes.

• Identifying specific DNA fragments/ molecules.

Principle

a) Electrophoresis technique uses principle of electric field on charge particles i.e.

• +ve molecules move towards -ve charge

• -ve molecules move towards +ve charge

b) It utilizes gel medium for holding and separation of molecules, that's why also called gel electrophoresis.

Mechanism/ Working/ Procedure

a) Gel (agarose or polyacrylamide) is sandwiched between glass or plastic plates to form a viscous slab.

b) Samples are placed on gel in special depressions called wells.

c) Two ends of the slab are suspended in two salt containing buffer solutions.

d) Buffer solutions are connected by electrodes to a power source.

e) Voltage is applied that moves molecules according to their charge;

• -ve molecules move to positive pole (electrode).

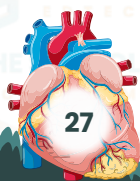
• +ve molecules move to negative pole (electrode).

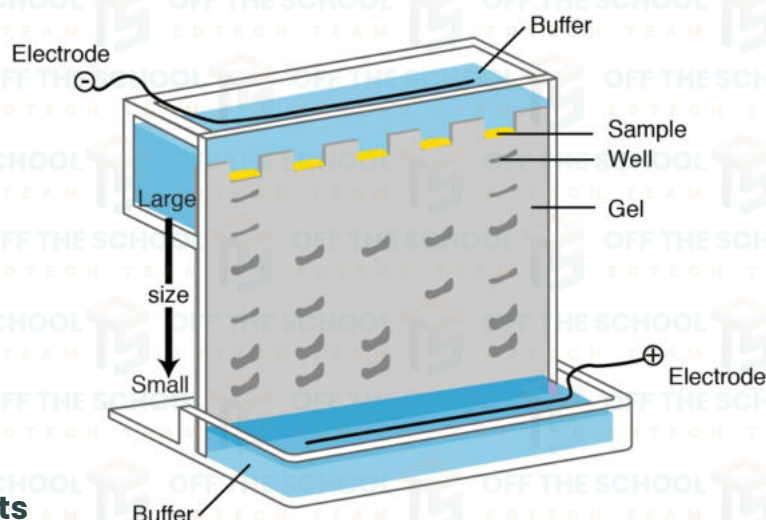
• Speed of molecules is inversely proportional to their size(mass).

• Smaller and lighter molecules move faster than larger and heavier molecules.

f) After the process, molecules are visualized using stains.

On the basis of Charge i.e. +ve or -ve , Shape ,Size ,Molecular weight.





Advanced Concepts

➤ Role of Gel

- Gel is porous_ so it allows molecules of to pass through different types of gels have different pore sizes.
- Pore size of a gel is also affected by its concentration i.e. at higher concentration (denser) gel forms small pores used for separation of small molecules.

➤ Role of buffer solution.

- It acts as electrolyte i.e. allows to pass current.

SPECTROPHOTOMETRY

- Quantitative technique
- Measures absorption of different wavelengths of light
- By a compound (mostly colored, but can also be used for colorless compounds)

Instrument:

- Measures the amount of light passing from sample.
- Calculates how much light is absorbed by sample.
- Can also find how much light is transmitted.

Principle

Every compound (based on structure and bonding) absorbs light of specific wavelengths.

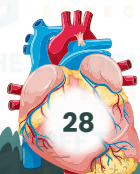
Result:

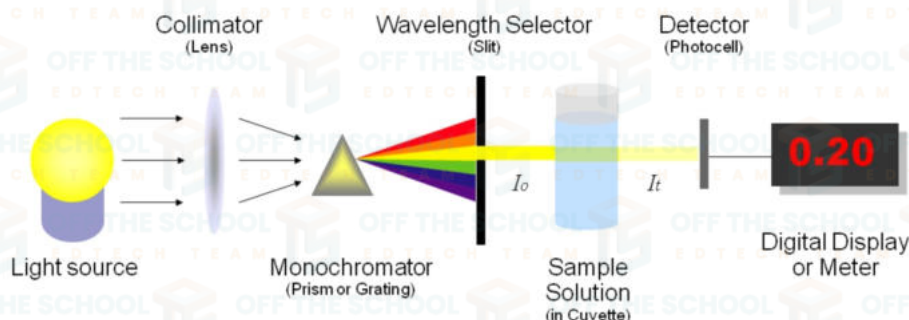
Absorption spectrum:

- It is a graph plotted by spectrophotometer
- Shows absorption of different light wavelengths by a compound

Structure and Working

Working bulb emits light which is focused by lens on monochromator. Specific wavelength is selected by wavelength selector and is passed from a sample solution in cuvette. Some part of light is absorbed by sample and remaining is transmitted. Transmitted light is detected by light detector and data is shown by a digital metre.





Applications

- Widely used in biology, chemistry, biochemistry, physics, chemical engineering, chemical industries
- Is used in diagnosis of diseases
- Can be used to find amount of a substance in samples.

MICROSCOPY

The study of cell and micro-organisms is dependent on microscope. The attributes of microscope are;

1. Magnification
2. Resolution

Magnification

- Ability of microscope to make enlarged image of an object.
- It is increase apparent size of object.
- It means how many times image of an object has been enlarged as compared to the actual size of the object.

Factors;

- It depends on magnification of lenses used in microscope.
- $M_m = M_o \times M_e$ Where;
- M_m = magnification of microscope.
- M_o ; magnification of objective lens..
- M_e = magnification of eyepiece (ocular lens).
- e.g. if M_o is 5X and M_e is then M_m will be 50X.

Resolution or resolving power

- It is the minimum distance at which two points can be seen separated.
- It is the measure of clarity of an image Units.
- Resolution is measured in mm.

MICROMETRY

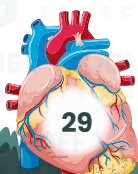
It is the measurement of size of microscopic objects using microscope.

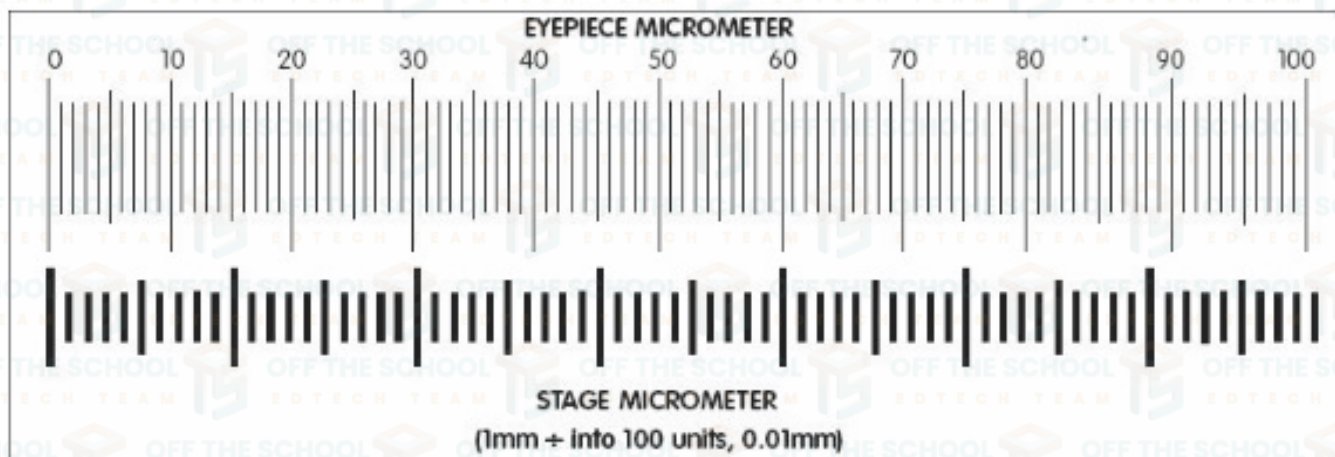
Principle

Calibration of ocular micrometer using stage micrometer.

Requirements;

- a) Slide of object
- b) Ocular micrometer
- c) Stage micrometer





Steps:

- Place slide of the object on stage of microscope
- Place ocular micrometer in eyepiece /ocular lens
- Note down divisions of ocular micrometer equal to length of the object
- Calibrate ocular micrometer with stage micrometer
- Remove slide of the object and place stage micrometer on the stage
- Join ends of two micrometers on one side
- Find divisions of stage micrometer equal to the divisions of ocular

CELL WALL

The cell wall is the outer most covering of a plant cell. It is composed of cellulose (a carbohydrate) and some other chemical substances. This hard covering gives form, firmness and strength to the plant cell. In a young cell it is thin and delicate but in a mature cell it becomes thick due to the deposition of various chemical substances on its inner surface.

There are three **layer of cell wall**.

1. Middle (lamella)

First formed cell plate.

Cementing layer between two daughter cells.

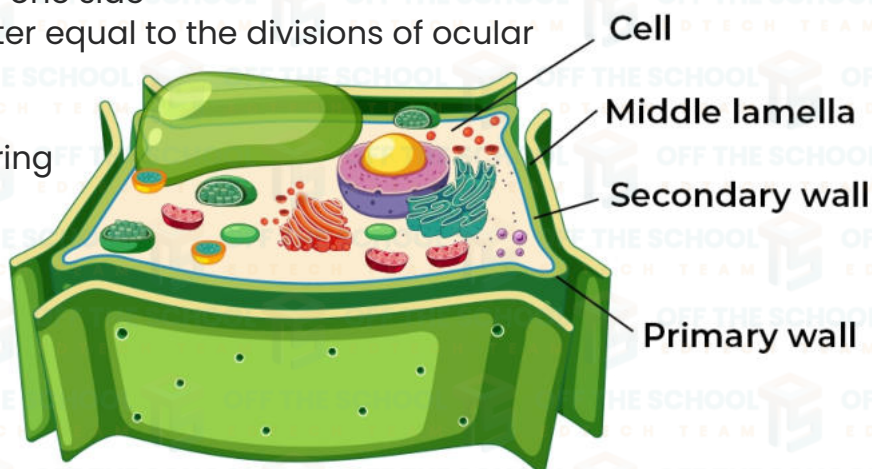
Composed of Ca^{++} and Mg^{++} pectate. Cells are separated when this layer is dissolved.

2. Primary wall

First product of cell synthesized by protoplast.

young cells it is thin and elastic while it becomes thick and rigid on maturity.

Made up of Hemicellulose (50%), cellulose (25%) and pectate substances.



PLASMA MEMBRANE (CELL MEMBRANE)

Plasma membrane or cell membrane is the outermost boundary of the animal cells and inner to the cell wall in plant cells. The cell membrane is chemically composed of lipids (20–40%) and proteins (60– 80%). In addition, there is a small number of carbohydrates present.

Fluid Mosaic Model

According to this model, there is a lipid bilayer in which the protein molecules are embedded. The fluid mosaic model is the most acceptable model of the plasma membrane. Many biologists contributed to establishing the structural organization of the cell membrane. The modern technology has revealed that lipid bilayer is not sandwiched between two protein layers. In 1972, Singer and Nicholson proposed a most acceptable model for membranes called the Fluid Mosaic Model. This model is in agreement with the photograph of the cell membrane by an electron microscope. This model explains that “the membrane is like a sea of lipids in which protein are floating”.

a. Proteins in the plasma membrane:

The proteins are not arranged in a sheet but as globes of proteins which are floating about in the sea of lipid molecules.

Intrinsic proteins:

Some proteins extend completely through the double layer of lipids from one side to the other and are called intrinsic proteins.

Extrinsic protein:

Some other proteins are smaller and are placed between the phospholipids molecules. These are on one side of the membrane and are called extrinsic proteins.

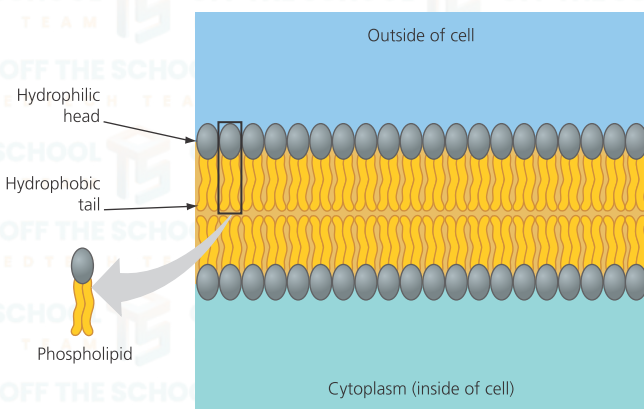
Role of extrinsic protein as receptor:

Extrinsic proteins function as a receptor that receives the stimuli from the environment and thus inform the cell to respond.

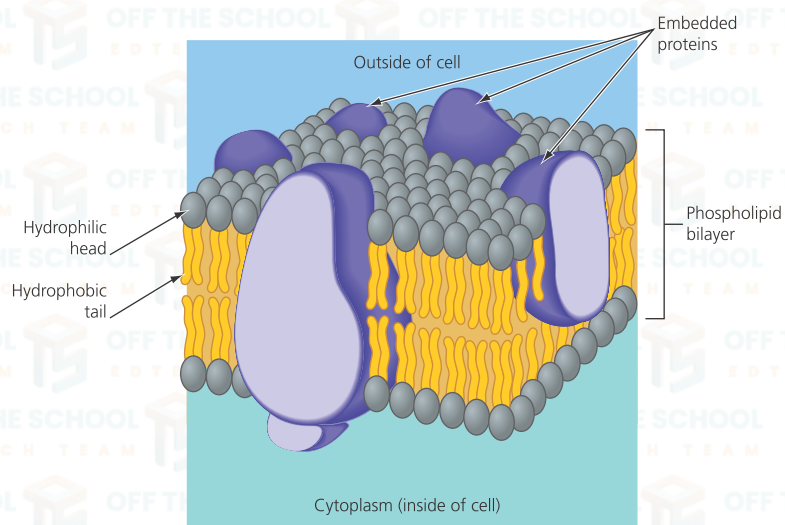
Functions of integral proteins:

Integral proteins also called permeases to regulate diffusion, osmosis and active transport of ionic materials.

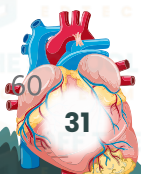
▼ The plasma membrane structure.



(a) Phospholipid bilayer of membrane. In water, phospholipids arrange themselves into a bilayer. The symbol for a phospholipid used in this book looks like a lollipop with two wavy sticks. The “head” is the end with the phosphate group, and the two “tails” are chains of carbon and hydrogen. The bilayer arrangement keeps the heads exposed to water while keeping the tails in the oily interior of the membrane.



(b) Fluid mosaic model of membrane. Membrane proteins, like the phospholipids, have both hydrophilic and hydrophobic regions.

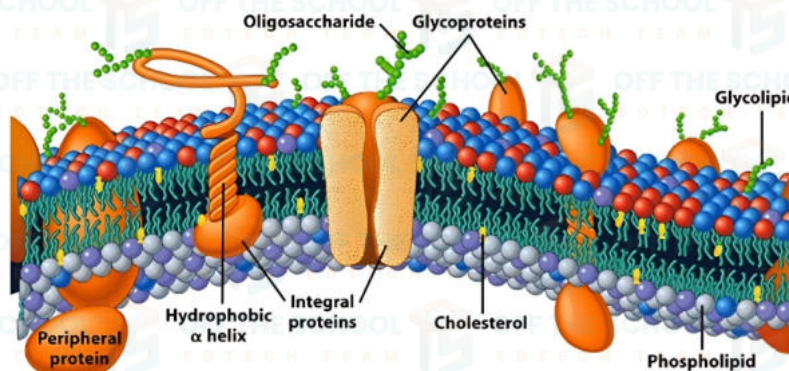


b. Glycolipids and glycoprotein in the plasma membrane:

Lipids, proteins, and carbohydrates are responsible for the functional diversity of the membranes. Lipid bilayer makes the membrane differentially permeable barrier that allows the transport of non-polar materials across it and prevents ionic materials. Membrane proteins, on the other hand, makes it a selectively permeable barrier that select materials according to the cell's need. If glucose concentration inside the cell is proper, no more glucose can enter the cell.

Functions of glycolipids and glycoprotein:

Membrane carbohydrates such as glycolipids and glycoproteins provide receptor sites that receive different types of stimuli like hormone receptor sites (HRS), receives nerve impulses, recognition of antigen and food materials thus inform the cell for a response. Membrane carbohydrates are also responsible for endocytosis i.e. phagocytosis (eating of the cell) and pinocytosis (drinking of the cell). Thus glycolipids and glycoproteins act as cell surface markers.



ENDOPLASMIC RETICULUM

The endoplasmic reticulum (ER) is a large organelle made of membranous sheets and tubules that begin near the nucleus and extend across the cell. The endoplasmic reticulum creates, packages, and secretes many of the products created by a cell.

Rough Endoplasmic Reticulum Structure

The rough endoplasmic reticulum is named so because of its appearance.

It is a series of connected flattened sacs having several ribosomes on its outer surface, hence the name.

It synthesizes and secretes proteins in the liver, hormones and other substances in the glands.

Rough ER is prominent in cells where protein synthesis happens (such as hepatocytes).

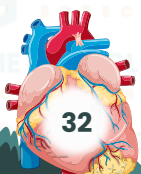
Smooth Endoplasmic Reticulum Structure

The smooth endoplasmic reticulum, on the other hand, does not have ribosomes.

The smooth endoplasmic reticulum has a tubular form.

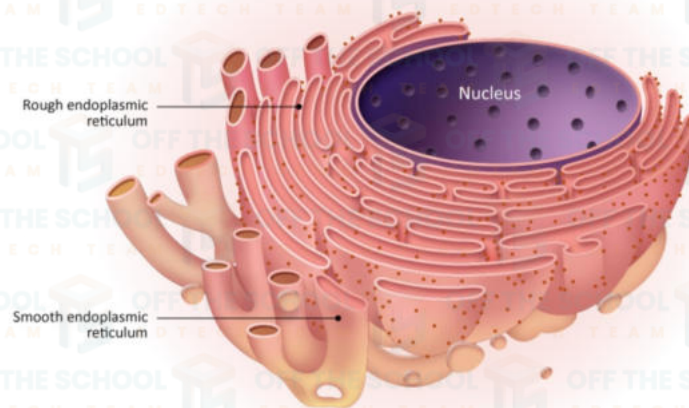
It participates in the production of phospholipids, the chief lipids in cell membranes and are essential in the process of metabolism.

Smooth ER transports the products of the rough ER to other cellular organelles.



Functions

- It is primarily in charge of transporting proteins and other carbohydrates to other organs, such as lysosomes, the Golgi apparatus, the plasma membrane, and so on.
- They also provide increased surface area for cellular reactions.
- They play an important role in the formation of the skeletal structure.
- Proteins, lipids, glycogen, and other steroid hormones such as cholesterol, progesterone, and testosterone require them for synthesis.
- They aid in the formation of nuclear membranes during cell division.
- The ER performs a variety of functions, including material transport between different parts of the cell.
- It eliminates waste and cleanses the cells. They also degrade worn-out cell components.
- by digesting foreign or undesirable materials and on a
- The membrane-bound sacs contain digestive enzymes that break down foreign material that enters the cell.

**RIBOSOMES**

Ribosomes are small, rounded, dense particles of ribonucleoprotein. They occur either freely in the cytoplasm and the matrix of mitochondria and chloroplast or remain attached with the membranes of the endoplasmic reticulum and nucleus. They are the sites of protein synthesis in both prokaryotic and eukaryotic cells.

Structures of Ribosomes

They are seen scattered in the cytoplasm and a few are connected to the endoplasmic reticulum.

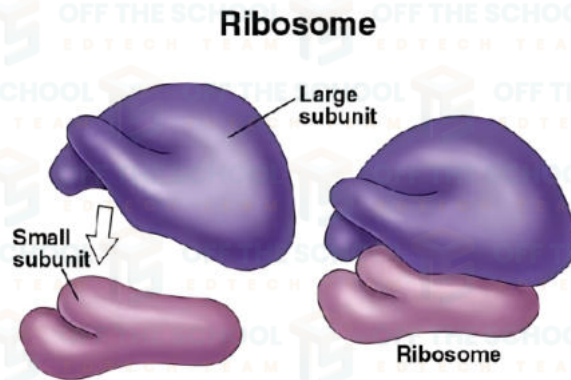
Whenever joined to the ER they are called the rough endoplasmic reticulum.

Around 37 to 62% of RNA is comprised of RNA and the rest is proteins.

Prokaryotes have 70S ribosomes respectively subunits comprising the little subunit of 30S and the bigger subunit of 50S.

Eukaryotes have 80S ribosomes respectively comprising of little (40S) and substantial (60S) subunits.

The ribosomes seen in the chloroplasts of mitochondria of eukaryotes are comprised of big and little subunits composed of proteins inside a 70S particle.



GOLGI COMPLEX

This apparatus which was found virtually in all eukaryotes consists of stacks of flattened, membrane-bounded sacs called cisternae. These cisternae together with associated vesicles are called Golgi-complex. It is a complex system of interconnected tubules around the central stack..

Cisternae:

Each dictyosome is formed of bundles of curved and flattened cisternae, associated tubules and secretory vesicles.

Dictyosome has two distinct faces.

The proximal (cis) or forming face present close to the nucleus.

A distal (trans) or maturing face located towards the cell membrane.

Vesicles and tubules pinched off from RER, flow, converge and fuse with the forming face to form new cisternae.

Functions of Golgi Apparatus:

- Golgi apparatus helps in the storage of secretory products and in the modification and packaging of secretory products.
- In some cases, polysaccharides may be synthesized from simple sugars in the Golgi apparatus. These polysaccharides may then be attached to proteins and lipids to form glycoproteins and glycolipids.
- Secretory vesicles produced by the Golgi apparatus may play an important role in adding surface area to the plasma membrane.

LYSOSOMES

Lysosomes are cytoplasmic organelles containing degradative enzymes enclosed in a membrane, found in most eukaryotes and are different from others cytoplasmic organelles due to their morphology."

Phagocytosis in lysosomes:

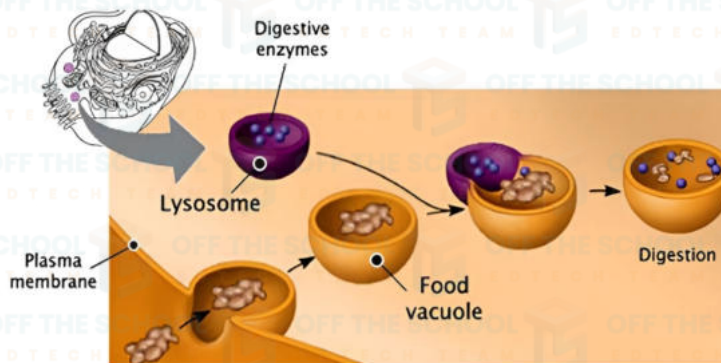
Any foreign objects that gain entry within the cell are immediately engulfed by the lysosomes and are completely broken down into digestible pieces. This process is known as phagocytosis. They are very abundant in those animal cells, which exhibit phagocytic activity e.g neutrophils.



Primary lysosomes:

Lysosomes are bounded by a single membrane and contain numerous hydrolytic and acidphosphatase enzymes. These enzymes are synthesized on RER and are further processed in the Golgi apparatus. The processed enzymes are budded off as Golgi vesicles and are called primary lysosomes.

These contain those enzymes which can digest the phagocytosed food particles. During autophagy (self-eating) some old worn out parts of the cell are also digested. In this way materials of the cell may be recycled and a cell may be renewed. Their enzymes can also result in degeneration of cell, as may occur during the developmental processes.



Malfunctioning of Lysosomes:

Several congenital diseases have been found to be due to the accumulation of substances such as glycogen or various lipids within the cell. These are also called storage diseases and are produced by a mutation that affects one of the lysosomal enzymes involved in the catabolism of certain substances.

Example:

In glycogenosis type II disease, the liver and muscles appear filled with glycogen within membrane-bound organelles. In this disease, an enzyme that converts glycogen to glucose is absent.

Tay-Sachs disease is involved in the catabolism of lipids. Accumulation of lipids in brain cells leads to mental retardation and even death.

PEROXISOMES

Structure

Peroxisomes are small, membrane-enclosed organelles that contain enzymes involved in a variety of metabolic reactions, including several aspects of energy metabolism. Or The peroxisomes are small sub-cellular bodies approximately 0.5µm in diameter, surrounded by membranes and found in a great variety of organisms, including plants and animals which are also-called "**microbodies.**"

In animals, they are most common in liver and kidney cells.

Function

- The enzymes of peroxisomes are known to break down hydrogen peroxide into oxygen and water, protecting cells from.
- In the leaves of green plants, it converts glycolate into amino acid and glycine by enzyme Glycolic acid oxidase during photorespiration.

GLYOXYSOMES

Glyoxysomes main function is to convert fatty acid to carbohydrate. They are present in plants and fungi.

- They are prevalent in the germinating seeds in their fat-storing tissues.

MITOCHONDRIA**Structure of Mitochondria**

Mitochondria are bound by a double membrane.

Outer Membrane

The outer membrane contains the enzymes involved in various activities. This contains pores called Porins which are permeable to macromolecules, ions, ATP, ADP, etc.

Inner Membrane

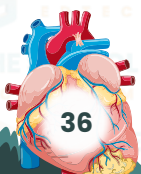
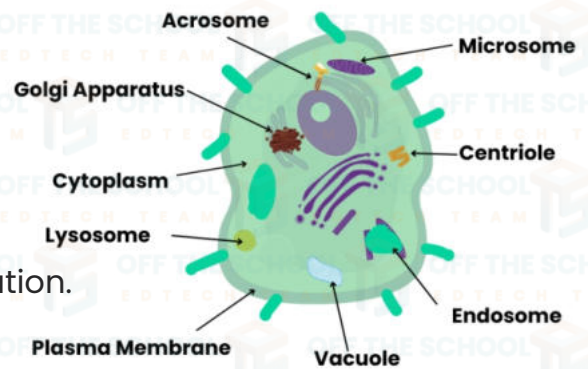
The inner mitochondrial membrane has several folds in it called Cristae, to increase the surface area. The inner membrane of the mitochondria is made up of proteins such as ATPase complex, cytochrome complex, NAD, FAD etc. catalyze the production of ATP. The inner mitochondrial membrane is permeable to oxygen, carbon dioxide, and water.

Matrix

It is fluidic contains proteins, ribosomes, enzymes, tRNA, and genomic DNA. Due to the presence of genetic material, the Mitochondrial Matrix can synthesize its RNA and proteins.

Functions

- The most important function of mitochondria is to produce energy. Mitochondria produce the molecule adenosine triphosphate (ATP).
- Mitochondria may also produce heat (brown fat), and accumulate iron-containing pigments (Heme ferritin), ions of Ca^{2+} and HPO_4^{2-} (or phosphate; e.g., osteoblasts of bones or yolk proteins).
- Mitochondria help the cells to maintain the proper concentration of calcium ions within the compartments of the cell.
- The mitochondria also help in building certain parts of blood and hormones like testosterone and estrogen.
- The liver cell's mitochondria have enzymes that detoxify ammonia.
- The mitochondria also play an important role in the process of apoptosis or programmed cell death.
- Abnormal death of cells due to the dysfunction of mitochondria can affect the function of an organ.



DNA and Inheritance in Mitochondria

Mitochondria replicate their DNA by a process called binary fission and can use this to make multiple copies in one mitochondrion.

Their DNA has **maternal lineage** which means their DNA is passed from mother to child with little change.

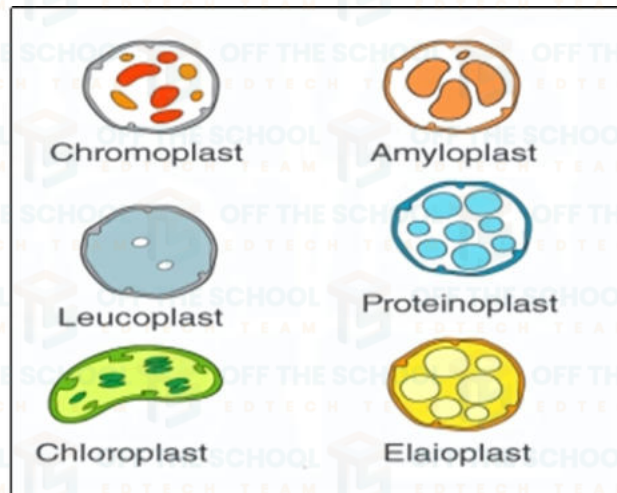
PLASTIDS

Plastids are double-membrane organelles which are found in the cells of plants and algae. Plastids are responsible for manufacturing and storing of food. These often contain pigments

Types of Plastids

There are different types of plastids with their specialized functions. Chloroplasts

1. Chromoplasts
2. Leucoplasts
3. Chloroplasts



Chromoplasts

These are found in flowering plants, ageing leaves and fruits. Chloroplasts convert into chromoplasts. Chromoplasts have carotenoid and Xanthophyll pigments that give different colours in leaves and fruits.

Leucoplasts

These are the non-pigmented or colourless plastids. Leucoplasts are found in storage parts of plant like roots to store starches, lipids, and proteins.

Leucoplasts are of three types:

Amyloplasts – Amyloplasts store and synthesize starch.

Proteinoplasts – Proteinoplasts help in storing the proteins in seeds.

Elaioplasts – Elaioplast helps in storing fats and oils.

Chloroplast

It is oval or biconvex, found within the mesophyll of the plant cell. They are double-membrane organelle. There are two distinct regions present inside a chloroplast known as the grana and stroma.

The chloroplast structure consists of the following parts:

1. Membrane Envelope

It comprises inner and outer lipid bilayer membranes. The inner membrane separates the stroma from the intermembrane space.

2. Intermembrane Space

The space between inner and outer membranes.

3. Thylakoid System (Lamellae)

The system is suspended in the stroma. It is a collection of membranous sacs called thylakoids or lamellae. The green coloured pigments called chlorophyll are found in the thylakoid membranes. It is the sight for the process of light-dependent reactions of the photosynthesis process. The thylakoids are arranged in stacks known as grana and each granum contains around 10-20 thylakoids.

4. Stroma

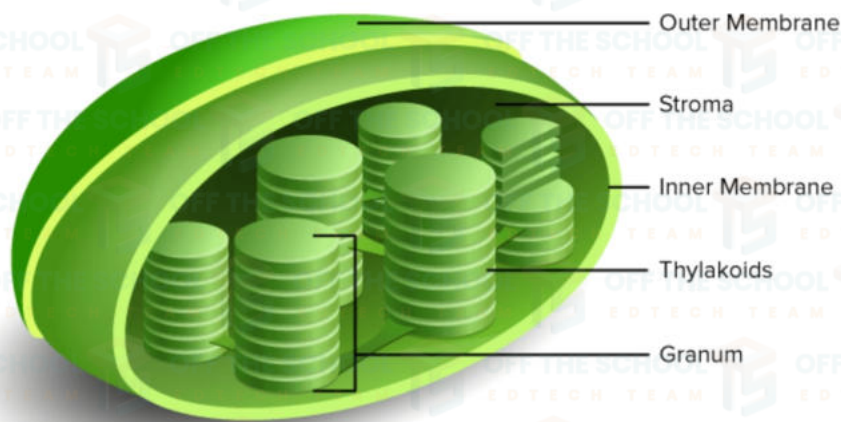
It is a colourless, alkaline, aqueous, protein-rich fluid present within the inner membrane of the chloroplast present surrounding the grana. Stroma also contains various enzymes, DNA, ribosomes, and other substances. Stroma lamellae connect the stacks of thylakoid sacs or grana.

5. Grana

Grana are made up of stacks of disc-shaped structures known as thylakoids or lamellae. The grana of the chloroplast consist of chlorophyll pigments. These are the sites of conversion of light energy into chemical energy.

6. Chlorophyll

It is a green photosynthetic pigment that helps in the process of photosynthesis.

**CYTOSKELETON**

The cytoskeleton is a structure that helps the cells to maintain their shape and internal organization, and it also provides mechanical support that enables cells to carry out essential functions like division and movement.”

Proteins present in cytoskeleton

The main proteins that are present in the cytoskeleton are tubulin (in microtubules), actin, myosin, tropomyosin, and others which are also found in muscles. Several cell organelles are derived from special assemblies of microtubules, e.g. cilia, flagella, basal bodies, and centrioles.

1. Microtubules:

These are long, unbranched, slender tubulin protein structures.

Functions of microtubules:

One very important function of microtubules is their role in the assembly and disassembly of the spindle structures during cell division.

2. Microfilaments:

These are considerably more thin cylinders made up of contractile actin proteins, linked to the inner face of the plasma membrane.

Function of microfilaments:

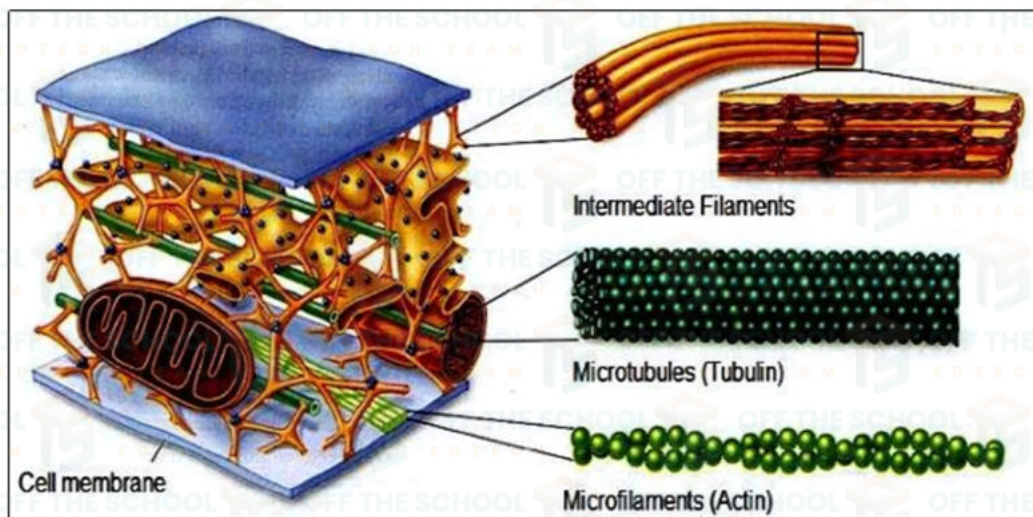
They are involved in cyclosis and amoeboid movements

3. Intermediate filaments:

They have a diameter in between those of microtubules and microfilaments.

Function of microfilaments:

They help in maintaining the cell shape.

**NUCLEUS****Nuclear membrane**

The nuclear envelope (also called a nuclear membrane) is the membrane that surrounds the nucleus. The nuclear membrane is also lipid bilayer. The nuclear membrane function is to regulate the entry and exit of materials.

Nucleoplasm

The nucleoplasm is the fluid filled within the nucleus. The nucleoplasm contains various materials e.g. chromosomes, nuclear bodies, and nuclear matrix.

Nuclear pores

They are small channels present within in the nuclear membrane. They control what molecules can go in or out.

Nucleolus

The nucleolus is as one of the nuclear bodies and is most prominent. It has round granular appearance. Its function is the synthesis of ribosomes.

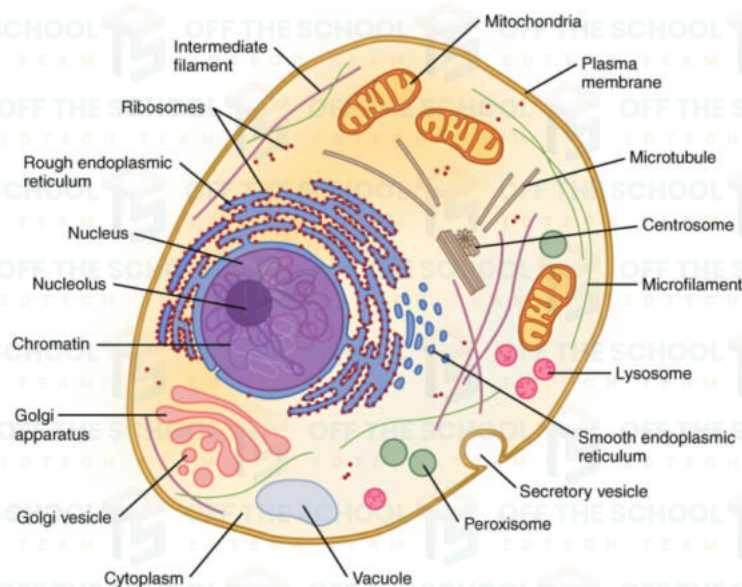
Nuclear matrix

It is a network of fiber that provides structural support to the size and shape of the nucleus.

Nucleus Function

Following are the important nucleus function:

- The nucleus is called the control center of the cell .It contains the cell's hereditary information and controls the cell's growth and reproduction.
- The nucleus controls the cell's activities such as metabolism, growth, and reproduction. It does so by regulating gene expression.
- It also mediates the replication of DNA and the Transcription of RNA.

**CHROMOSOMES****Centromere**

It is also known as Kinetochore and is the primary constriction at the center where the chromatids or spindle fibers are attached. It functions in the movement of the chromosome during a stage called anaphase during cell division.

Chromatid

When a chromosome is divided into two identical strands during cell division, a chromatid is formed as half of the chromosome. Each half strand is joined by a centromere, both are known as sister chromatids; and it contains DNA and separates at Anaphase to form a separate chromosome.

Chromatin

Chromatin is a complex of DNA consisting of DNA, RNA, and protein and it forms chromosomes within the nucleus of eukaryotic cells. It doesn't exist freely as linear stands, rather it is highly condensed and wrapped around nuclear proteins.

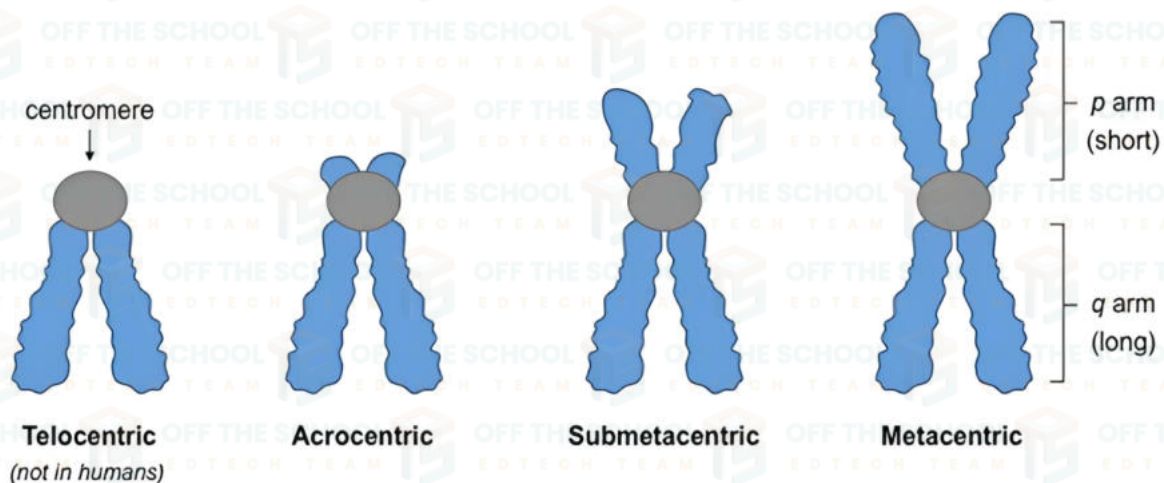
Telomere

The terminal region of each side of the chromosome is called a telomere.

Types of Chromosomes

1. **Telocentric** are rod-shaped chromosomes with centromere occupying the terminal position, so that the chromosome has just one arm.
2. **Acrocentric** are also rod-shaped chromosomes with centromere occupying a sub-terminal position. One arm is very long and the other is very short.
3. **Sub-metacentric** chromosomes are with centromere slightly away from the mid-point so that the two arms are unequal.
4. **Metacentric** are V-shaped chromosomes in which centromere lies in the middle of chromosome so that the two arms are almost equal.

On the Basis of Number of Centromeres



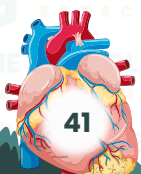
VACUOLE

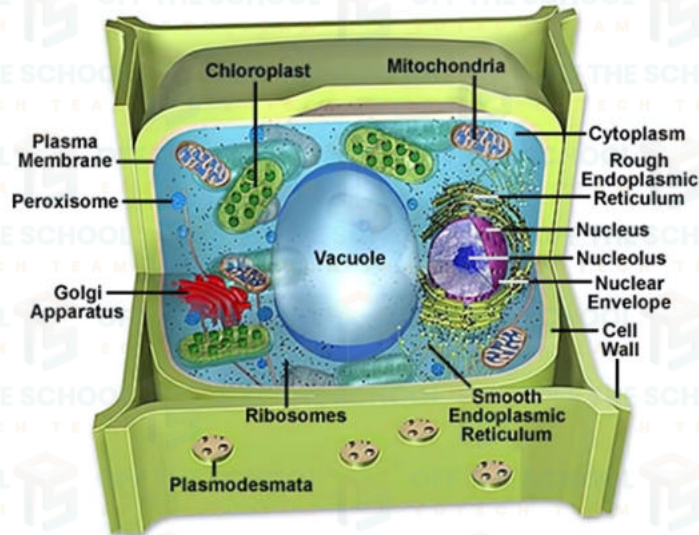
a) In Young Plant Cells

- In young plant cells, many small vacuoles are present
- These small vacuoles;
 - Can hold reserves of important organic compounds
 - In some plants they can store poisonous or unpleasant for protection against herbivores.

b) In Mature Plant Cells

- Mature plant cells have large vacuoles
- Large vacuoles are formed by fusion of small vacuoles
- Present at the centre of cell therefore also called central vacuole
- Membrane of the central vacuole is called tonoplast
- The solution inside the central vacuole is called cell sap.
 - Is main reservoir of inorganic ions e.g K^+ , Cl^- etc.
 - Acts as storehouse of the plant cell
 - Provides mechanical support by developing turgor pressure.





CENTRIOLES

Centrioles are non-membranous cell organelles that form spindle fibres during cell division

- They occur mainly in animal cells
- Also present in some fungi-like organisms e.g slime molds and water molds
- Absent in higher plants most fungi and prokaryotes
- Centrioles are rod shaped structures
- Centrioles lie in a distinctly staining region of cytoplasm called centrosphere
- They are usually present in pairs
- They occur at right angle to each other near one pole of nucleus
- Centrioles and centrosphere are collectively called centrosome

Structure

- Each centriole consists of 9 triplets of microtubules (triplet: set of three; $9 \times 3 = 27$ microtubules)
- Triplets of microtubules are circularly arranged around a central axis

Functions

- Just before division, two centrioles duplicate and make two pairs
- Pairs of centrioles migrate to opposite sides of nucleus
- Pairs of centrioles form spindle fibres during cell division (prophase)
- The whole set of spindle fibres is called mitotic apparatus which helps in distribution of chromosomes.

